

Statistical & Distributional Analysis

1. Overview

This report summarizes the statistical exploratory data analysis (EDA) performed across four related datasets describing state and non-state cyber conflict incidents. The analysis focuses primarily on the incident-level dataset (`global_dataset_clean.csv`), with supporting analysis of dyadic initiator–receiver relationships to examine how incident severity and impact vary by actor type and operation type.

The datasets analyzed are:

Dataset	Rows	Columns	Description
<code>global_dataset_clean.csv</code>	3,414	84	One row per incident (main table)
<code>dyadic_dataset_clean.csv</code>	4,296	51	One row per initiator–receiver dyad
<code>attribution_dataset_clean.csv</code>	5,188	16	One row per attribution claim
<code>receiver_dataset_clean.csv</code>	12,180	6	One row per receiver (victim) entity

Of the 84 columns in the global dataset, the majority (72) are text/categorical fields, 10 are integer counts, 1 is boolean, and 1 is a continuous float variable, reflecting a dataset that is largely descriptive/categorical with a smaller set of quantitative severity and impact metrics.

2. Missingness Overview

Many fields in the global dataset are conditionally populated — for example, legal-response fields are only completed when a legal response actually occurred. As a result, high missingness in specific columns is often expected rather than indicative of a data-quality problem. Missingness was nonetheless profiled across all columns prior to any statistical testing.

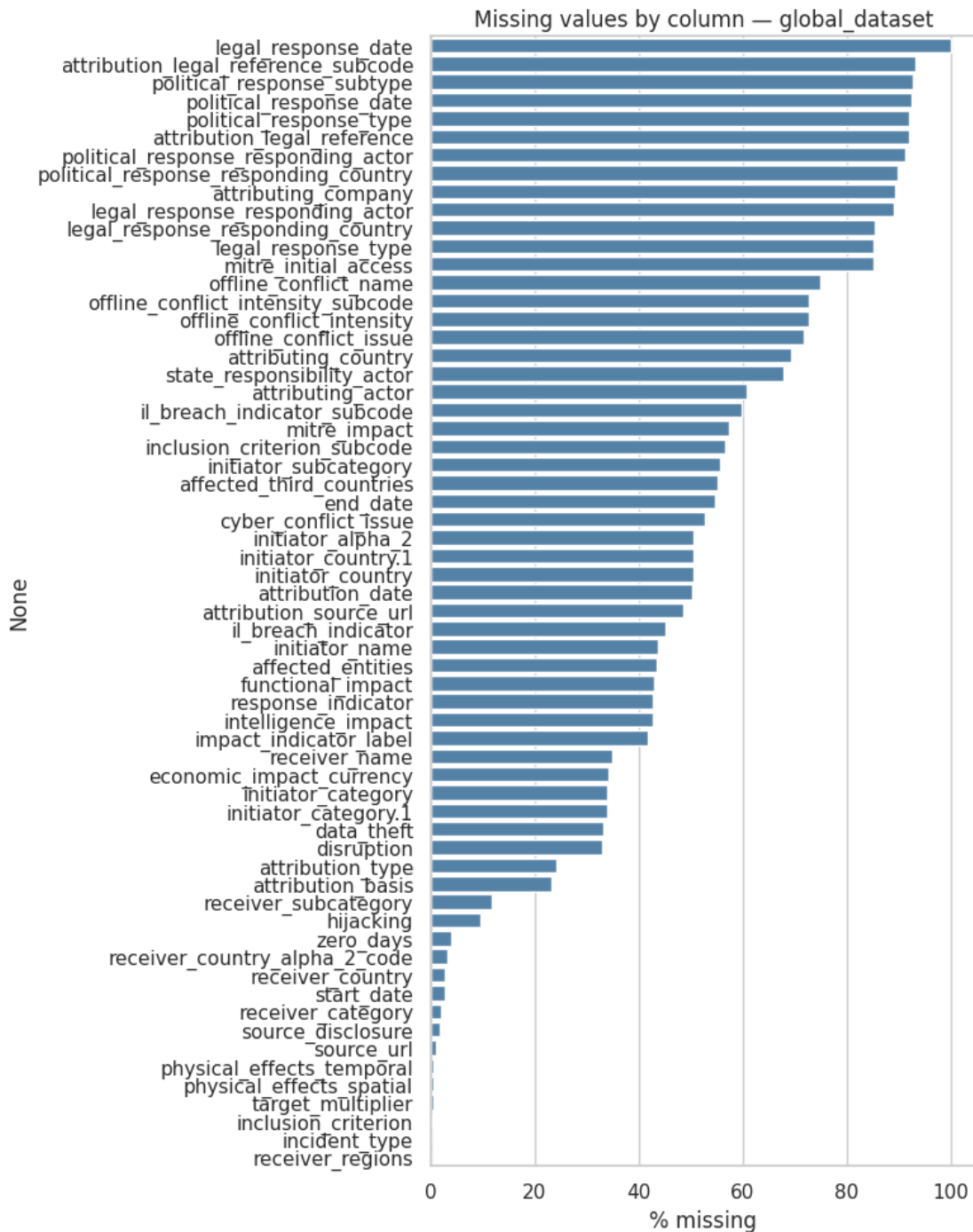


Figure 1. Percentage of missing values by column, global_dataset_clean.csv.

Fields such as legal_response_date show substantial missingness because most incidents never trigger a legal response — the absence of data reflects the absence of the underlying event, not a collection error.

3. Descriptive Statistics (Numeric Variables)

Nine numeric variables were retained for quantitative summary after excluding identifier and date-like columns. Descriptive statistics (mean, standard deviation, and quartiles) are shown below.

Variable	Mean	Std Dev	Min	25%	Median	75%	Max
Number of attributions	1.14	0.51	0	1	1	1	10
Number of political responses	0.11	0.50	0	0	0	0	13
Unweighted intensity	2.43	1.35	0	1	2	3	7
Weighted intensity	2.44	1.43	0	1	2	3	11
Impact indicator score	3.94	3.61	0	0	5	7	15
Affected entities value	297.6	17,115.3	0	0	0	1	1,000,000
Affected third countries value	0.48	0.88	0	0	0	1	23
Economic impact value (\$)	857,210.9	19,455,740	0	0	0	0	872,000,000
Number of legal responses	0.15	0.39	0	0	0	0	3

The large gap between the mean and median for economic_impact_value and affected_entities_value (means far exceeding medians of 0) indicates strong right-skew: most incidents involve little to no measurable economic loss or entity impact, but a small number of extreme incidents pull the mean up dramatically — consistent with a heavy-tailed, "black swan" incident pattern typical of cyber conflict data.

4. Distributions

Histograms with kernel density estimates were used to examine the shape of the core severity and response metrics.

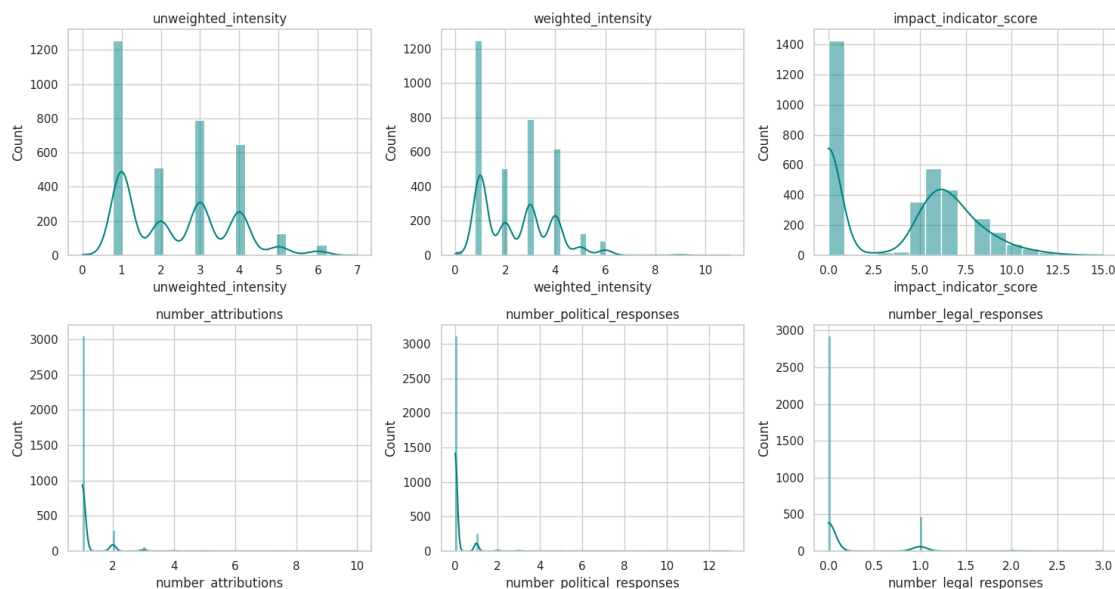


Figure 2. Distributions of intensity, impact, and response-count variables.

Most incidents cluster at low intensity and impact scores, with counts of attributions, political responses, and legal responses concentrated near zero — the large majority of incidents receive a single attribution and no formal political or legal response.

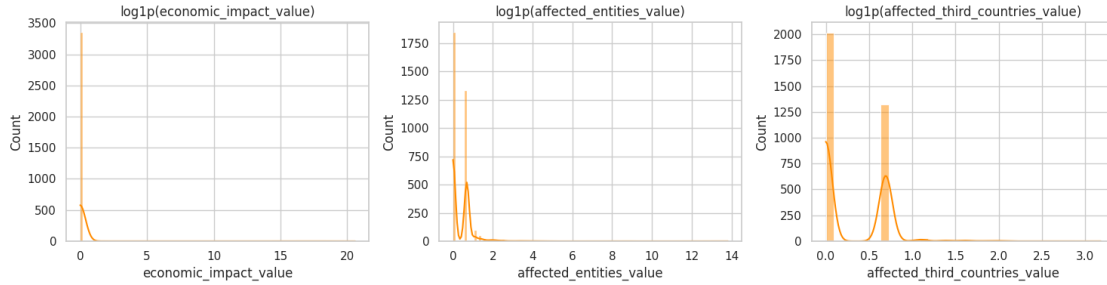


Figure 3. Log1p-transformed distributions of economic impact, affected entities, and affected third countries.

Because economic_impact_value and affected_entities_value are extremely right-skewed in raw form, a log1p transform was applied for visualization. This reveals that even after log-scaling, the bulk of incidents remain concentrated at the low end, with a long tail of substantially larger-scale events.

5. Outlier Detection

Outliers were identified using two complementary methods: the 1.5×IQR rule and a Z-score threshold of $|Z| > 3$. Boxplots were also used for visual inspection.

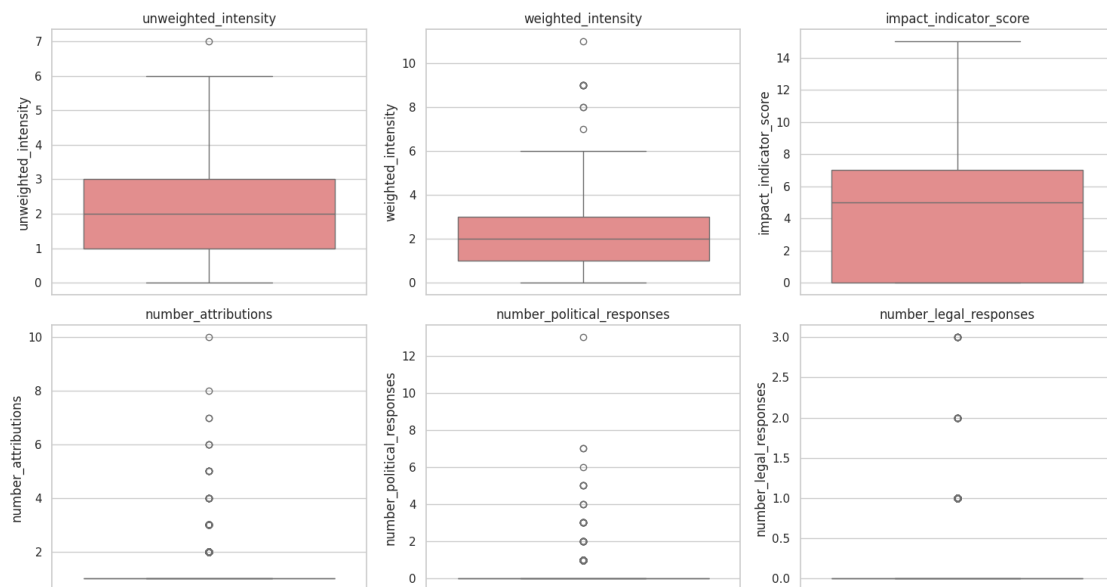


Figure 4. Boxplots of intensity, impact, and response-count variables.

In this dataset, boxplot outliers largely represent unusually severe or high-response incidents — the statistical equivalent of "black swan" cyber events that are meaningfully more severe or drew significantly more attributions/responses than a typical incident, rather than data errors.

5.1 IQR-Based Outlier Counts

Variable	Outliers (IQR rule)	% of Records	Lower Bound	Upper Bound
Number of legal responses	488	14.29%	0.0	0.0
Number of attributions	366	10.72%	1.0	1.0
Number of political responses	297	8.70%	0.0	0.0
Affected entities value	151	4.42%	-1.5	2.5

Variable	Outliers (IQR rule)	% of Records	Lower Bound	Upper Bound
Economic impact value	70	2.05%	0.0	0.0
Affected third countries value	45	1.32%	-1.5	2.5
Weighted intensity	11	0.32%	-2.0	6.0
Unweighted intensity	1	0.03%	-2.0	6.0
Impact indicator score	0	0.00%	-10.5	17.5

5.2 Z-Score-Based Outlier Counts ($|Z| > 3$)

Variable	Outliers ($ Z > 3$)
Number of attributions	72
Number of political responses	40
Affected third countries value	32
Number of legal responses	28
Weighted intensity	11
Economic impact value	9
Impact indicator score	1
Unweighted intensity	1
Affected entities value	1

number_legal_responses (14.3%) and number_attributions (10.7%) show the highest outlier rates under the IQR rule, reflecting that while most incidents receive exactly one attribution and no legal response, a meaningful minority involve multiple attributions or legal actions. impact_indicator_score shows zero IQR outliers, indicating a comparatively well-behaved, non-extreme distribution relative to the other metrics.

6. Relationships Between Variables

6.1 Scatterplots

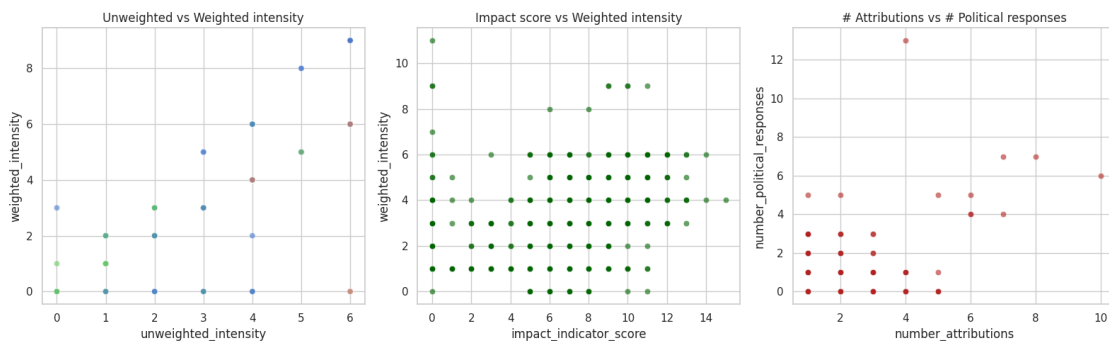


Figure 5. Pairwise relationships between intensity, impact, and response metrics.

Unweighted and weighted intensity move together almost linearly, as expected since weighted intensity is a refinement of the unweighted measure. The relationship between number of attributions and number of political responses is examined to assess whether states respond more assertively when attacker attribution is more certain.

6.2 Pairplot by Initiator Category



Figure 6. Pairwise distributions of severity and response metrics, colored by initiator category.

Separating the severity/response metrics by initiator_category (e.g., state vs. non-state actors) allows a visual comparison of whether different types of actors produce systematically different incident profiles. The diagonal panels show each variable's distribution split by actor type, while the off-diagonal panels show pairwise scatter relationships.

7. Dyadic-Level Relationship Checks

The dyadic dataset — one row per initiator–receiver pair within an incident — was used to test whether incident severity and impact differ systematically by initiator category and by operation type.

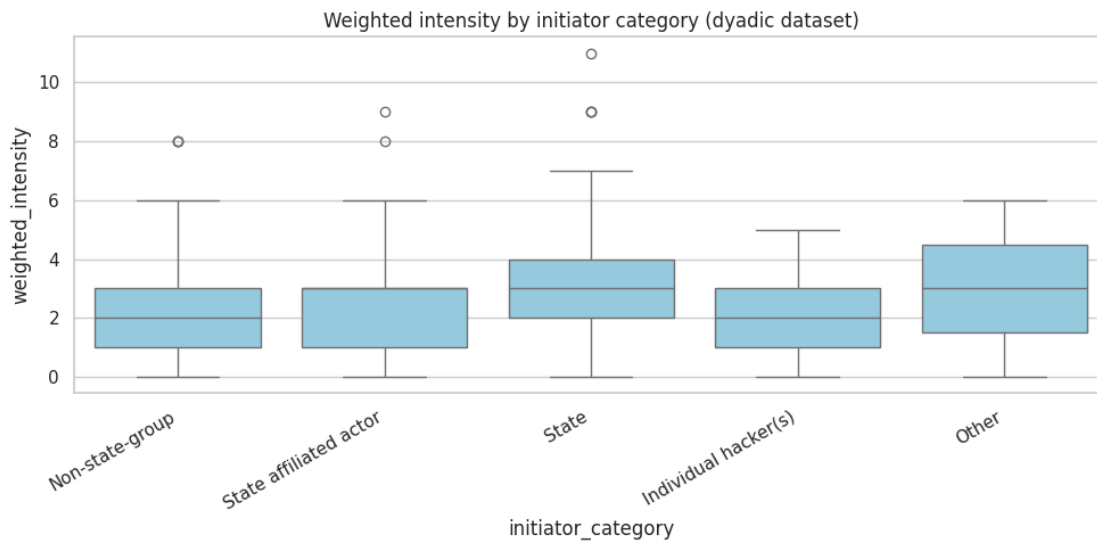


Figure 7. Weighted intensity by initiator category.

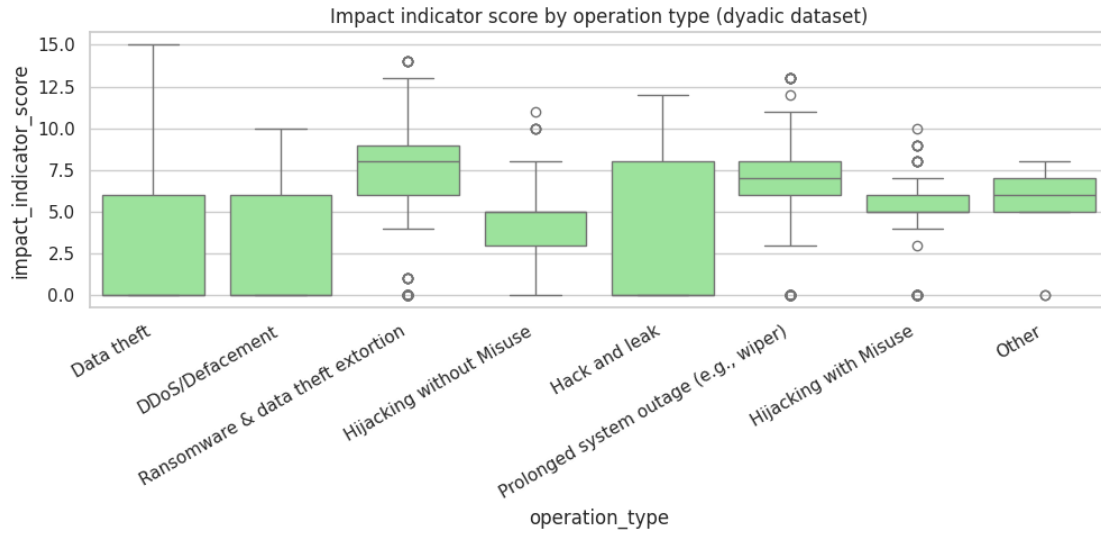


Figure 8. Impact indicator score by operation type.

Ransomware and data-theft/extortion operations tend to show a higher median impact indicator score than simpler operation types such as basic data theft, consistent with the disruptive and coercive nature of ransomware and extortion campaigns.

7.1 One-Way ANOVA: Impact by Initiator Category

A one-way ANOVA was used to formally test whether impact_indicator_score differs significantly across initiator_category groups.

Test	F-statistic	p-value	Conclusion
One-way ANOVA (impact_indicator_score ~ initiator_category)	54.415	< 0.0001	Statistically significant difference between groups

With $F = 54.415$ and $p < 0.0001$, the null hypothesis of equal mean impact scores across initiator categories is rejected at any conventional significance level. This provides strong statistical evidence that the type of initiating actor (e.g., state-sponsored vs. non-state) is associated with meaningfully different incident impact levels.

8. Key Findings & Summary

- **Distributional shape:** Severity and response metrics (intensity, impact, attribution/response counts) are concentrated at low values, with a small number of high-severity "black swan" incidents driving the tails of the distributions.
- **Skewed impact:** economic_impact_value and affected_entities_value are extremely right-skewed even after log-transformation, indicating that a small number of incidents account for a disproportionate share of total economic and entity-level impact.
- **Outliers:** number_legal_responses and number_attributions have the highest outlier rates (14.3% and 10.7% under the IQR rule), while impact_indicator_score shows no IQR-flagged outliers, suggesting it is a comparatively stable, well-behaved composite measure.

- **Strong relationship:** Unweighted and weighted intensity are almost perfectly correlated, as expected given their construction.
- **Operation type matters:** Ransomware and data-theft/extortion operations are associated with higher impact indicator scores than simpler operation types.
- **Initiator category is significant:** Impact indicator score differs significantly across initiator categories (one-way ANOVA, $F = 54.415$, $p < 0.0001$), indicating that who initiates an incident is statistically associated with how severe its impact is.
- **Data quality caveat:** High missingness in fields like legal_response_date most likely reflects the conditional structure of the data (the underlying event did not occur) rather than data-quality issues, and should be handled accordingly in downstream modeling.

Overall, the exploratory analysis indicates that cyber conflict incidents follow a heavy-tailed severity distribution dominated by many low-impact events and a minority of extreme incidents, and that both the actor initiating an incident and the type of operation used are meaningfully associated with its resulting impact.